Fast Document Retrieval Engine for a University RAG Assistant

Short Report with Program Output: CS-251 Data Structures and Algorithms (25CS)
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Subject teacher: Dr. Ali Asghar Manjotho · **Submitted:** [date]

The full technical report is `docs/REPORT.md`. Every screenshot below was taken from our program on 26 Oct 2026 (Windows 11, g++ 10.3 `-O2`). The data is set in MUET's CSE department but, apart from D24 (the PBL brief), every rule is **synthetic** and not official MUET policy.

1. What we built

The **retrieval** step of a RAG assistant. Given a question, the program returns the top-k passages with their source (passage id, document, version, date, CURRENT/SUPERSEDED status). If there is not enough support, it answers **"No supported evidence found."**

- **Data:** 31 documents, 155 passages, 5 topics; 20 labelled questions (8 keyword, 4 paraphrase, 5 version/ambiguous, 3 unanswerable).
- **Baseline:** a linear scan that reads every token of every passage, O(T·q).
- **Index:** our own **hash table** (separate chaining, FNV-1a, resize at load 0.75) maps each word to a **postings list** `{passage, tf}`. A search only visits the postings of the question's words, O(q + P).
- **Ranking:** BM25 + freshness (superseded passages × 0.5), or simple coverage. **Evidence rule:** at least 50 % of the question's concepts must match.
- **Top-k:** our own bounded **min-heap**, O(C log k); merge sort, O(C log C), for comparison.
- **Front ends:** the command-line program `rag.exe` and a desktop GUI that calls it.

2. Key results

Measure	Result
Recall@3 (155 passages, 17 answerable questions)	**0.912** (0.794 without synonyms)
Unanswerable questions refused	**3 / 3**
Scan vs index, 155 passages	16.75 µs vs 1.49 µs (about **11×**)
Scan vs index, 15,500 passages	1,900 µs vs 130 µs (about **15×**)
Index memory, 155 passages	64 KB
Heap vs merge sort (C = 9,700, k = 3)	0.68 ms vs 2.54 ms, same top-k
Scan and index give identical results	YES, every question, ranker and size
Tests	**24 / 24 pass**

3. Program output: command line

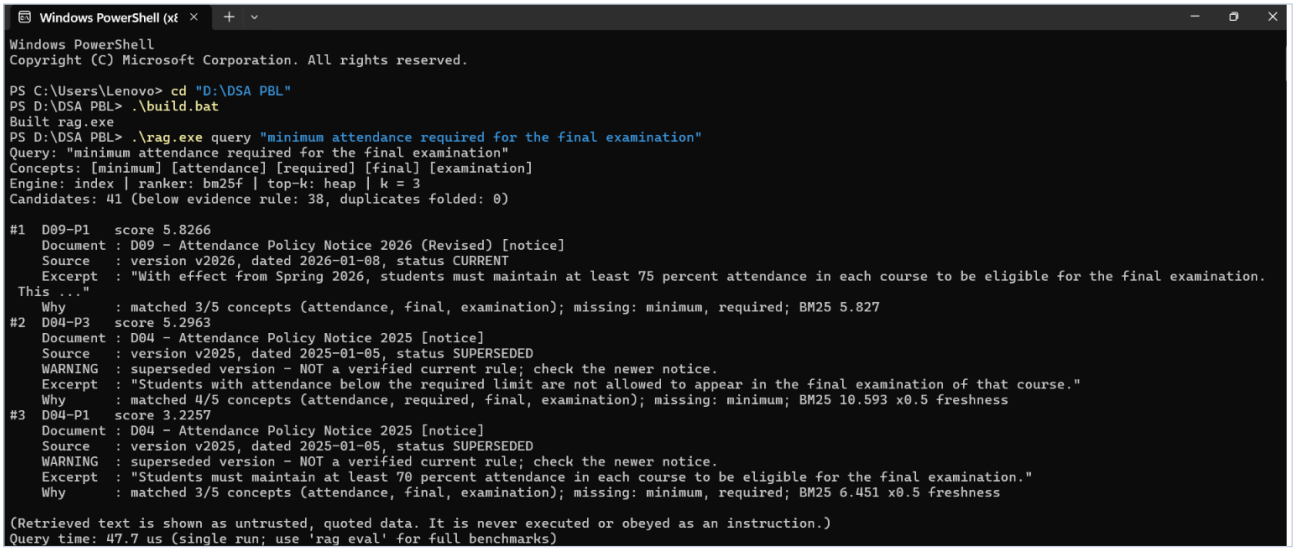


Figure 1. *Build and a normal query. The current 2026 attendance notice (75 %) ranks first; the old 2025 notice matches more words (BM25 10.593) but is SUPERSEDED, so the freshness rule halves its score and it drops below.*

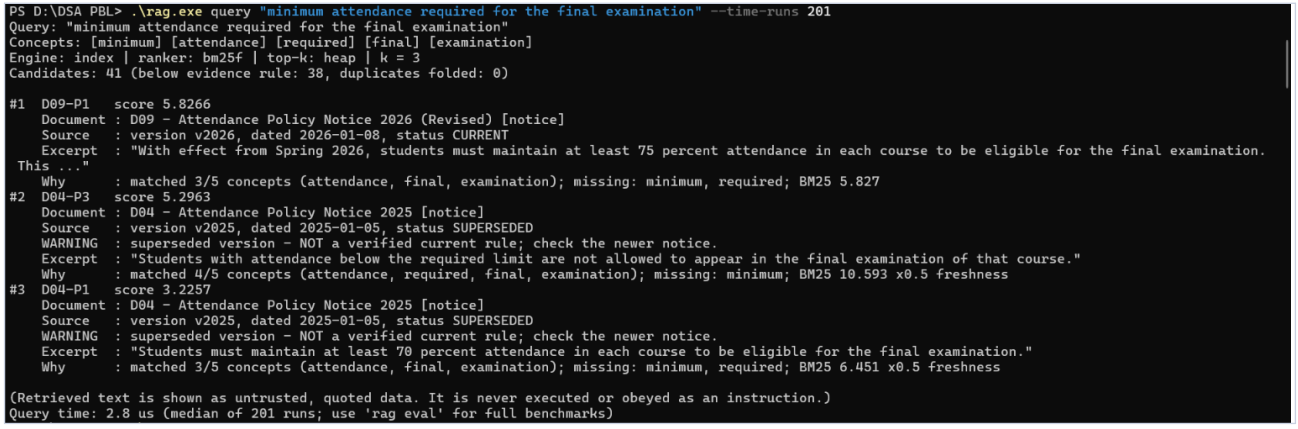


Figure 2. *The same query timed over 201 runs: median 2.8 µs. A single first run (47.7 µs above) is slower because the program has just started.*

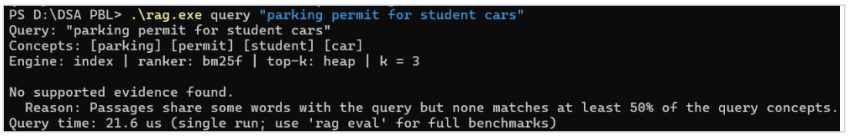


Figure 3. *Unanswerable question. Passages share the word "student", but none matches 50 % of the concepts, so the program refuses instead of inventing a rule.*

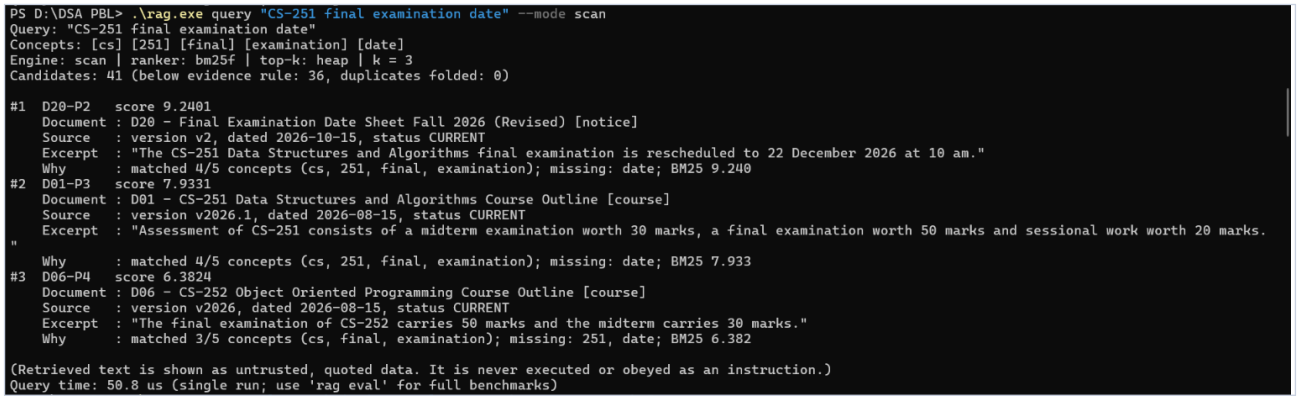


Figure 4. *Linear-scan baseline (--mode scan). The revised date sheet D20-P2 is returned first, with its source and the reason for its score.*

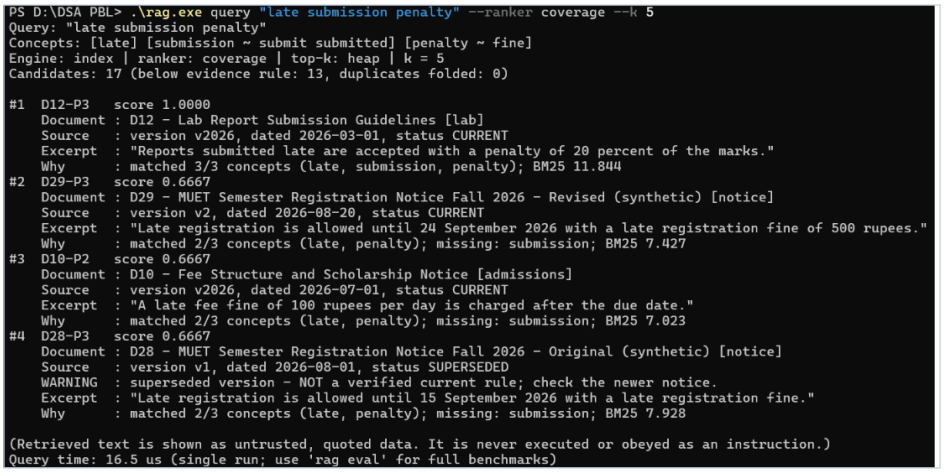


Figure 5. *Coverage ranker with k = 5 and synonym expansion (submission ~ submit, penalty ~ fine). The superseded D28 notice is flagged with a warning.*

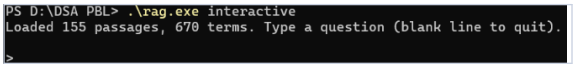


Figure 6. *Interactive mode: 155 passages and 670 unique terms loaded.*

4. Program output: tests (`rag test`)

```
PS D:\DSA PBL> .\rag.exe test
Edge-case and validation tests (24 cases)
================================================================
T01  [PASS]  Empty corpus
      expected: No supported evidence (corpus is empty), both engines
      observed: The corpus is empty. / scan: The corpus is empty.
T02  [PASS]  Blank / stop-word-only query
      expected: No supported evidence; query empty after preprocessing
      observed: The query is empty after preprocessing (blank or only stop words). | stop words only: no evidence
T03  [PASS]  Term not present
      expected: No supported evidence, 0 candidates
      observed: no evidence, candidates 0
T04  [PASS]  Repeated query terms
      expected: Collapse to 1 concept; identical ranking to single term
      observed: 1 concept(s); D09-P5, D09-P1, D13-P4
T05  [PASS]  Duplicate term occurrences in a passage
      expected: One posting per passage with tf = 2
      observed: D09-P1 'attendance': 1 posting, tf=2; D21-P4 'fee': 1 posting, tf=2
T06  [PASS]  Multiple matching passages
      expected: Many candidates, exactly k = 3 returned, scores descending
      observed: 15 candidates -> D02-P1, D17-P3, D23-P4
T07  [PASS]  Exact duplicate passages (D02-P1 / D23-P1)
      expected: Original shown once, repost folded
      observed: D02-P1, D23-P4, D02-P3; folded 1
T08  [PASS]  Near-duplicate (unauthorised/unauthorized)
      expected: Not detected as duplicate (known limitation)
      observed: D23-P3 duplicate of D02-P3? no
T09  [PASS]  Hash collisions (4 buckets, 200 keys, no resize)
      expected: All 200 keys found with correct values
      observed: max chain 50, colliding entries 196
T10  [PASS]  Worst-case hash (all keys same bucket)
      expected: Chaining still correct for find/erase/insert
      observed: used buckets 1, longest chain 50
T11  [PASS]  Index on degenerate hash (V terms in 1 chain)
      expected: Same results as normal index for all 20 labelled queries
      observed: identical: yes, longest chain 670
T12  [PASS]  Old/new notice (attendance 2025 vs 2026)
      expected: 2026 notice ranked above 2025; 2025 marked SUPERSEDED
      observed: D09-P1, D04-P3, D04-P1
T13  [PASS]  Old/new date sheet (D19 vs D20)
```

Figure 7. *Tests T01-T13: empty corpus, blank query, missing word, repeated words, duplicates, hash collisions (4 buckets / 200 keys; all keys in one bucket; whole index in one chain of 670) and old/new notices.*

```
T14  [PASS]  Coverage ranker on version-sensitive query
      expected: Superseded D04 passage can outrank current one
      observed: D04-P3, D09-P1, D04-P1
T15  [PASS]  Scan vs index consistency
      expected: Identical ranked ids and scores for every labelled query, both rankers
      observed: 40/40 identical
T16  [PASS]  Heap top-k vs merge sort top-k
      expected: Same top-k for k = 1, 3, 10 on every labelled query
      observed: 60/60 identical
T17  [PASS]  k = 50 (> evidence) and k = 0
      expected: Returns only qualifying passages; k = 0 returns none
      observed: 2 hits for k=50; k=0 -> k = 0: no passages requested.
T18  [PASS]  Unanswerable questions
      expected: No supported evidence found (no invented rule)
      observed: parking: no evidence (46 weak candidates); cafeteria: no evidence
T19  [PASS]  Prompt-injection text in a passage (D22-P3)
      expected: Returned only as quoted data; later answers unchanged
      observed: injection query -> D22-P3; attendance query still -> D09-P1, D04-P3, D04-P1
T20  [PASS]  Add / update / remove after indexing
      expected: New passage found; old text gone after update; nothing after removal
      observed: added -> TEST-P1; old text -> no evidence; new text -> TEST-P1; removed -> no evidence
T21  [PASS]  Duplicate passage_id
      expected: Rejected at ingestion
      observed: rejected
T22  [PASS]  Tie-break on equal scores
      expected: Newer date first, then file order; repeat runs identical
      observed: D25-P2, D06-P2, D23-P4
T23  [PASS]  Heap and merge sort unit check
      expected: top-3 = 9,9,7; sorted = 1 2 3 5 7 9 9
      observed: heap top-3 9,9,7; sorted 1 2 3 5 7 9 9
================================================================
24/24 passed
```

Figure 8. *Tests T14-T23: scan = index (40/40), heap = merge sort (60/60), k limits, unanswerable questions, injection text, add/update/remove, duplicate id, tie-break. Result: 24/24 passed.*

5. Program output: evaluation (`rag eval`)

```
PS D:\DSA PBL> .\rag.exe eval
======================================================================
 Fast Document Retrieval Engine - evaluation report
======================================================================
Corpus file  : data/passages.txt
Queries file : data/queries.txt (20 questions)
k = 3, evidence rule: >= 50% of query concepts matched
Latency: 5 warm-up runs, then 15 measured runs of 40 back-to-back searches; median reported.

----------------------------------------------------------------------
Corpus size: 25 passages, 284 tokens (T), 197 unique terms (V), 271 postings
Index build time (median of 7): 0.048 ms
Index memory: buckets 4.0 KB + nodes 12.3 KB + long keys 0.0 KB + postings 2.3 KB = 18.6 KB
Document storage (not index): text+metadata 12.6 KB, token arrays 12.5 KB
Hash table: 512 buckets, load 0.38, longest chain 3, colliding entries 32, resizes 3

Q    type         ranker    index top-3                 R@3     scan us    index us   same?
Q01  keyword      coverage  D02-P1 D02-P3               1.00    3.69       0.91       yes
Q02  keyword      coverage  D03-P2 D03-P1               1.00    3.28       0.82       yes
Q03  keyword      coverage  NO-EVIDENCE                 n/a     3.30       0.48       yes
Q04  keyword      coverage  NO-EVIDENCE                 n/a     2.16       0.34       yes
Q05  keyword      coverage  D01-P2                      1.00    3.73       0.72       yes
Q06  paraphrase   coverage  D02-P1 D02-P3               1.00    3.80       0.72       yes
Q07  paraphrase   coverage  NO-EVIDENCE                 n/a     3.56       0.64       yes
Q08  paraphrase   coverage  NO-EVIDENCE                 n/a     2.30       0.20       yes
Q09  version      coverage  D04-P3 D04-P1               n/a     3.61       0.95       yes
Q10  version      coverage  D01-P3                      n/a     3.38       0.85       yes
Q11  ambiguous    coverage  D01-P3 D03-P4 D04-P1        1.00    3.12       1.09       yes
Q12  ambiguous    coverage  NO-EVIDENCE                 0.00    2.80       0.40       yes
Q13  unanswerable coverage  NO-EVIDENCE                 ok      2.65       0.50       yes
Q14  unanswerable coverage  NO-EVIDENCE                 ok      1.42       0.16       yes
Q15  keyword      coverage  NO-EVIDENCE                 n/a     1.42       0.16       yes
Q16  keyword      coverage  NO-EVIDENCE                 n/a     2.22       0.30       yes
Q17  paraphrase   coverage  NO-EVIDENCE                 n/a     4.06       0.76       yes
Q18  version      coverage  NO-EVIDENCE                 n/a     3.21       0.55       yes
Q19  keyword      coverage  NO-EVIDENCE                 n/a     1.23       0.15       yes
Q20  unanswerable coverage  NO-EVIDENCE                 ok      1.65       0.17       yes
>> coverage @ 25 passages: average Recall@3 = 0.833 over 6 answerable questions (with gold in subset); unanswerable correctly refused 3/3; median query time
 scan 3.17 us vs index 0.52 us
```

Figure 9. *Evaluation at 25 passages, coverage ranker: index memory 18.6 KB; Recall@3 0.833; unanswerable refused 3/3; scan 3.17 µs vs index 0.52 µs.*

```
Q01  keyword      bm25f     D02-P1 D02-P3               1.00    2.79       0.68       yes
Q02  keyword      bm25f     D03-P2 D03-P1               1.00    2.92       0.74       yes
Q03  keyword      bm25f     NO-EVIDENCE                 n/a     2.92       0.47       yes
Q04  keyword      bm25f     NO-EVIDENCE                 n/a     2.07       0.33       yes
Q05  keyword      bm25f     D01-P2                      1.00    3.60       0.71       yes
Q06  paraphrase   bm25f     D02-P1 D02-P3               1.00    3.72       0.73       yes
Q07  paraphrase   bm25f     NO-EVIDENCE                 n/a     3.47       0.64       yes
Q08  paraphrase   bm25f     NO-EVIDENCE                 n/a     2.30       0.20       yes
Q09  version      bm25f     D04-P3 D04-P1               n/a     3.61       0.97       yes
Q10  version      bm25f     D01-P3                      n/a     3.40       0.86       yes
Q11  ambiguous    bm25f     D01-P3 D03-P4 D04-P1        1.00    3.16       0.98       yes
Q12  ambiguous    bm25f     NO-EVIDENCE                 0.00    2.76       0.40       yes
Q13  unanswerable bm25f     NO-EVIDENCE                 ok      2.67       0.50       yes
Q14  unanswerable bm25f     NO-EVIDENCE                 ok      1.44       0.16       yes
Q15  keyword      bm25f     NO-EVIDENCE                 n/a     1.41       0.16       yes
Q16  keyword      bm25f     NO-EVIDENCE                 n/a     2.23       0.30       yes
Q17  paraphrase   bm25f     NO-EVIDENCE                 n/a     4.04       0.76       yes
Q18  version      bm25f     NO-EVIDENCE                 n/a     3.22       0.54       yes
Q19  keyword      bm25f     NO-EVIDENCE                 n/a     1.22       0.14       yes
Q20  unanswerable bm25f     NO-EVIDENCE                 ok      1.66       0.17       yes
>> bm25f @ 25 passages: average Recall@3 = 0.833 over 6 answerable questions (with gold in subset); unanswerable correctly refused 3/3; median query time sc
an 2.85 us vs index 0.52 us

----------------------------------------------------------------------
Corpus size: 50 passages, 524 tokens (T), 310 unique terms (V), 503 postings
Index build time (median of 7): 0.060 ms
Index memory: buckets 4.0 KB + nodes 19.4 KB + long keys 0.0 KB + postings 4.4 KB = 27.8 KB
Document storage (not index): text+metadata 24.9 KB, token arrays 23.0 KB
Hash table: 512 buckets, load 0.61, longest chain 4, colliding entries 75, resizes 3

Q    type         ranker    index top-3                 R@3     scan us    index us   same?
Q01  keyword      coverage  D02-P1 D02-P3               1.00    4.35       0.77       yes
Q02  keyword      coverage  D03-P2 D03-P1               1.00    4.38       0.87       yes
Q03  keyword      coverage  D10-P2 D10-P1               1.00    4.92       0.82       yes
Q04  keyword      coverage  D10-P3                      1.00    3.52       0.45       yes
Q05  keyword      coverage  D01-P2                      1.00    6.25       0.98       yes
Q06  paraphrase   coverage  D02-P1 D02-P3               1.00    6.23       0.90       yes
Q07  paraphrase   coverage  D08-P2 D08-P1               0.50    5.93       1.01       yes
Q08  paraphrase   coverage  NO-EVIDENCE                 n/a     4.79       0.33       yes
Q09  version      coverage  D04-P3 D09-P1 D04-P1        0.50    5.70       1.39       yes
```

Figure 10. *25 passages with BM25 + freshness, then 50 passages (coverage): every row shows "same? yes" (scan and index agree).*

```
Q09  version      coverage  D04-P3 D09-P1 D04-P1          0.50    5.70       1.39       yes
Q10  version      coverage  D01-P3 D06-P4                 n/a     5.66       1.32       yes
Q11  ambiguous    coverage  D01-P3 D06-P4 D03-P4          1.00    4.78       1.56       yes
Q12  ambiguous    coverage  D10-P2                        0.00    4.60       0.49       yes
Q13  unanswerable coverage  NO-EVIDENCE                   ok      4.52       0.78       yes
Q14  unanswerable coverage  NO-EVIDENCE                   ok      2.39       0.16       yes
Q15  keyword      coverage  NO-EVIDENCE                   n/a     2.35       0.16       yes
Q16  keyword      coverage  NO-EVIDENCE                   n/a     3.79       0.37       yes
Q17  paraphrase   coverage  NO-EVIDENCE                   n/a     7.23       1.19       yes
Q18  version      coverage  NO-EVIDENCE                   n/a     5.38       0.81       yes
Q19  keyword      coverage  NO-EVIDENCE                   n/a     1.96       0.14       yes
Q20  unanswerable coverage  NO-EVIDENCE                   ok      2.78       0.17       yes
>> coverage @ 50 passages: average Recall@3 = 0.800 over 10 answerable questions (with gold in subset); unanswerable correctly refused 3/3; median query tim
e scan 4.69 us vs index 0.79 us

Q01  keyword      bm25f     D02-P1 D02-P3                 1.00    4.25       0.78       yes
Q02  keyword      bm25f     D03-P2 D03-P1                 1.00    4.41       0.89       yes
Q03  keyword      bm25f     D10-P2 D10-P1                 1.00    4.96       0.84       yes
Q04  keyword      bm25f     D10-P3                        1.00    3.54       0.47       yes
Q05  keyword      bm25f     D01-P2                        1.00    6.30       0.99       yes
Q06  paraphrase   bm25f     D02-P1 D02-P3                 1.00    6.21       0.92       yes
Q07  paraphrase   bm25f     D08-P2 D08-P1                 0.50    5.87       1.03       yes
Q08  paraphrase   bm25f     NO-EVIDENCE                   n/a     4.81       0.32       yes
Q09  version      bm25f     D09-P1 D04-P3 D04-P1          0.50    5.77       1.39       yes
Q10  version      bm25f     D01-P3 D06-P4                 n/a     5.65       1.34       yes
Q11  ambiguous    bm25f     D06-P4 D01-P3 D03-P4          1.00    4.73       1.51       yes
Q12  ambiguous    bm25f     D10-P2                        0.00    4.55       0.51       yes
Q13  unanswerable bm25f     NO-EVIDENCE                   ok      4.53       0.78       yes
Q14  unanswerable bm25f     NO-EVIDENCE                   ok      2.35       0.16       yes
Q15  keyword      bm25f     NO-EVIDENCE                   n/a     2.35       0.16       yes
Q16  keyword      bm25f     NO-EVIDENCE                   n/a     3.80       0.37       yes
Q17  paraphrase   bm25f     NO-EVIDENCE                   n/a     7.21       1.18       yes
Q18  version      bm25f     NO-EVIDENCE                   n/a     5.35       0.81       yes
Q19  keyword      bm25f     NO-EVIDENCE                   n/a     1.96       0.14       yes
Q20  unanswerable bm25f     NO-EVIDENCE                   ok      2.85       0.17       yes
>> bm25f @ 50 passages: average Recall@3 = 0.800 over 10 answerable questions (with gold in subset); unanswerable correctly refused 3/3; median query time s
can 4.64 us vs index 0.79 us
```

Figure 11. *50 passages: Recall@3 0.800 for both rankers; scan 4.6 µs vs index 0.79 µs.*

```
Corpus size: 100 passages, 986 tokens (T), 491 unique terms (V), 955 postings
Index build time (median of 7): 0.130 ms
Index memory: buckets 8.0 KB + nodes 30.7 KB + long keys 0.0 KB + postings 8.5 KB = 47.2 KB
Document storage (not index): text+metadata 49.0 KB, token arrays 43.5 KB
Hash table: 1024 buckets, load 0.48, longest chain 4, colliding entries 112, resizes 4

Q    type         ranker    index top-3                   R@3     scan us    index us   same?
Q01  keyword      coverage  D02-P1 D02-P3                 1.00    8.39       1.08       yes
Q02  keyword      coverage  D03-P2 D03-P1                 1.00    9.25       1.96       yes
Q03  keyword      coverage  D10-P2 D10-P1                 1.00    10.90      1.68       yes
Q04  keyword      coverage  D10-P3                        1.00    6.88       0.57       yes
Q05  keyword      coverage  D01-P2                        1.00    13.44      1.31       yes
Q06  paraphrase   coverage  D02-P1 D02-P3                 1.00    13.11      1.60       yes
Q07  paraphrase   coverage  D08-P2 D08-P1                 0.50    15.81      1.90       yes
Q08  paraphrase   coverage  D14-P2                        1.00    11.47      0.54       yes
Q09  version      coverage  D04-P3 D09-P1 D04-P1          0.50    12.43      2.34       yes
Q10  version      coverage  D20-P2 D01-P3 D19-P2          1.00    11.54      2.58       yes
Q11  ambiguous    coverage  D01-P3 D06-P4 D20-P1          1.00    9.57       2.92       yes
Q12  ambiguous    coverage  D12-P3 D10-P2                 0.50    10.40      1.14       yes
Q13  unanswerable coverage  NO-EVIDENCE                   ok      9.12       1.21       yes
Q14  unanswerable coverage  NO-EVIDENCE                   ok      5.42       0.28       yes
Q15  keyword      coverage  NO-EVIDENCE                   n/a     4.46       0.38       yes
Q16  keyword      coverage  NO-EVIDENCE                   n/a     8.27       0.65       yes
Q17  paraphrase   coverage  NO-EVIDENCE                   n/a     15.43      2.10       yes
Q18  version      coverage  D20-P1                        n/a     10.90      1.32       yes
Q19  keyword      coverage  NO-EVIDENCE                   n/a     3.63       0.14       yes
Q20  unanswerable coverage  NO-EVIDENCE                   ok      6.20       0.17       yes
>> coverage @ 100 passages: average Recall@3 = 0.875 over 12 answerable questions (with gold in subset); unanswerable correctly refused 3/3; median query ti
me scan 9.98 us vs index 1.26 us

Q01  keyword      bm25f     D02-P1 D02-P3                 1.00    8.41       1.10       yes
Q02  keyword      bm25f     D03-P2 D03-P1                 1.00    9.39       1.99       yes
Q03  keyword      bm25f     D10-P2 D10-P1                 1.00    10.97      1.71       yes
Q04  keyword      bm25f     D10-P3                        1.00    6.90       0.58       yes
Q05  keyword      bm25f     D01-P2                        1.00    13.28      1.30       yes
Q06  paraphrase   bm25f     D02-P1 D02-P3                 1.00    13.18      1.62       yes
Q07  paraphrase   bm25f     D08-P2 D08-P1                 0.50    12.33      1.92       yes
Q08  paraphrase   bm25f     D14-P2                        1.00    11.58      0.55       yes
Q09  version      bm25f     D09-P1 D04-P3 D04-P1          0.50    12.27      2.34       yes
Q10  version      bm25f     D20-P2 D01-P3 D06-P4          1.00    11.70      2.55       yes
```

Figure 12. *100 passages: 1,024 buckets, longest chain 4; Recall@3 0.875; scan 9.98 µs vs index 1.26 µs.*

```
Q11  ambiguous    bm25f     D06-P4 D01-P3 D13-P3          1.00    9.36      2.60      yes
Q12  ambiguous    bm25f     D12-P3 D10-P2                 0.50    10.40     1.16      yes
Q13  unanswerable bm25f     NO-EVIDENCE                   ok      9.23      1.20      yes
Q14  unanswerable bm25f     NO-EVIDENCE                   ok      5.43      0.28      yes
Q15  keyword      bm25f     NO-EVIDENCE                   n/a     4.41      0.38      yes
Q16  keyword      bm25f     NO-EVIDENCE                   n/a     8.22      0.65      yes
Q17  paraphrase   bm25f     NO-EVIDENCE                   n/a     16.02     2.10      yes
Q18  version      bm25f     D20-P1                        n/a     10.89     1.33      yes
Q19  keyword      bm25f     NO-EVIDENCE                   n/a     3.64      0.14      yes
Q20  unanswerable bm25f     NO-EVIDENCE                   ok      6.24      0.17      yes
>> bm25f @ 100 passages: average Recall@3 = 0.875 over 12 answerable questions (with gold in subset); unanswerable correctly refused 3/3; median query time
scan 9.90 us vs index 1.25 us

------------------------------------------------------------------
Corpus size: 155 passages, 1557 tokens (T), 670 unique terms (V), 1501 postings
Index build time (median of 7): 0.176 ms
Index memory: buckets 8.0 KB + nodes 41.9 KB + long keys 0.0 KB + postings 14.1 KB = 64.0 KB
Document storage (not index): text+metadata 93.1 KB, token arrays 68.2 KB
Hash table: 1024 buckets, load 0.65, longest chain 4, colliding entries 181, resizes 4

Q    type         ranker    index top-3                   R@3     scan us    index us   same?
Q01  keyword      coverage  D02-P1 D23-P4 D23-P5          1.00    13.11      1.46       yes
Q02  keyword      coverage  D03-P2 D03-P1                 1.00    13.15      1.48       yes
Q03  keyword      coverage  D10-P2 D10-P1                 1.00    19.86      1.44       yes
Q04  keyword      coverage  D10-P3                        1.00    9.54       0.57       yes
Q05  keyword      coverage  D01-P2                        1.00    21.87      1.90       yes
Q06  paraphrase   coverage  D02-P1 D23-P3 D23-P4          1.00    24.96      2.29       yes
Q07  paraphrase   coverage  D08-P2 D08-P1                 0.50    18.05      1.62       yes
Q08  paraphrase   coverage  D14-P2                        1.00    17.68      0.58       yes
Q09  version      coverage  D04-P3 D09-P1 D04-P1          0.50    20.39      2.54       yes
Q10  version      coverage  D20-P2 D01-P3 D19-P2          1.00    19.37      2.88       yes
Q11  ambiguous    coverage  D01-P3 D06-P4 D20-P1          1.00    12.62      3.09       yes
Q12  ambiguous    coverage  D12-P3 D29-P3 D10-P2          0.50    20.94      1.30       yes
Q13  unanswerable coverage  NO-EVIDENCE                   ok      14.09      1.86       yes
Q14  unanswerable coverage  NO-EVIDENCE                   ok      8.52       0.28       yes
Q15  keyword      coverage  D27-P1                        1.00    8.60       0.43       yes
Q16  keyword      coverage  D25-P1                        1.00    12.92      1.05       yes
Q17  paraphrase   coverage  D26-P1                        1.00    28.70      3.24       yes
Q18  version      coverage  D29-P1 D28-P1 D20-P1          1.00    17.12      2.27       yes
Q19  keyword      coverage  D31-P1 D31-P3 D31-P4          1.00    7.50       0.71       yes
Q20  unanswerable coverage  NO-EVIDENCE                   ok      10.05      0.20       yes
```

Figure 13. *100 passages (BM25 + freshness) and the full 155-passage corpus: index memory 64.0 KB, load factor 0.65, longest chain 4.*

```
>> coverage @ 155 passages: average Recall@3 = 0.912 over 17 answerable questions (with gold in subset); unanswerable correctly refused 3/3; median query ti
me scan 15.61 us vs index 1.47 us

Q01  keyword      bm25f     D02-P1 D23-P4 D02-P3          1.00    13.07      1.40       yes
Q02  keyword      bm25f     D03-P2 D03-P1                 1.00    13.19      1.51       yes
Q03  keyword      bm25f     D10-P2 D10-P1                 1.00    16.23      1.47       yes
Q04  keyword      bm25f     D10-P3                        1.00    9.62       0.58       yes
Q05  keyword      bm25f     D01-P2                        1.00    21.99      1.91       yes
Q06  paraphrase   bm25f     D02-P1 D23-P3 D23-P4          1.00    22.13      2.23       yes
Q07  paraphrase   bm25f     D08-P2 D08-P1                 0.50    17.99      1.63       yes
Q08  paraphrase   bm25f     D14-P2                        1.00    20.64      0.59       yes
Q09  version      bm25f     D09-P1 D04-P3 D04-P1          0.50    20.43      2.60       yes
Q10  version      bm25f     D20-P2 D01-P3 D06-P4          1.00    19.45      2.85       yes
Q11  ambiguous    bm25f     D06-P4 D01-P3 D25-P5          1.00    15.78      2.86       yes
Q12  ambiguous    bm25f     D12-P3 D29-P3 D10-P2          0.50    22.27      1.26       yes
Q13  unanswerable bm25f     NO-EVIDENCE                   ok      18.43      1.89       yes
Q14  unanswerable bm25f     NO-EVIDENCE                   ok      8.62       0.28       yes
Q15  keyword      bm25f     D27-P1                        1.00    8.57       0.44       yes
Q16  keyword      bm25f     D25-P1                        1.00    12.88      1.06       yes
Q17  paraphrase   bm25f     D26-P1                        1.00    26.91      3.25       yes
Q18  version      bm25f     D29-P1 D20-P1 D28-P1          1.00    17.26      2.27       yes
Q19  keyword      bm25f     D31-P1 D31-P5 D31-P4          1.00    7.41       0.70       yes
Q20  unanswerable bm25f     NO-EVIDENCE                   ok      10.04      0.17       yes
>> bm25f @ 155 passages: average Recall@3 = 0.912 over 17 answerable questions (with gold in subset); unanswerable correctly refused 3/3; median query time
scan 16.75 us vs index 1.49 us

Scan and index returned identical ranked results for every query/ranker/size: YES

------------------------------------------------------------------
Ablation: synonym expansion OFF (full corpus, index engine)
>> coverage: average Recall@3 = 0.794 (17 answerable), unanswerable refused 3/3
   per query: Q01=1.00 Q02=1.00 Q03=1.00 Q04=1.00 Q05=1.00 Q06=0.00 Q07=0.50 Q08=0.00 Q09=0.50 Q10=1.00 Q11=1.00 Q12=0.50 Q15=1.00 Q16=1.00 Q17=1.00 Q18=1.0
0 Q19=1.00
>> bm25f: average Recall@3 = 0.794 (17 answerable), unanswerable refused 3/3
   per query: Q01=1.00 Q02=1.00 Q03=1.00 Q04=1.00 Q05=1.00 Q06=0.00 Q07=0.50 Q08=0.00 Q09=0.50 Q10=1.00 Q11=1.00 Q12=0.50 Q15=1.00 Q16=1.00 Q17=1.00 Q18=1.0
0 Q19=1.00
```

Figure 14. *155 passages: Recall@3 0.912 for both rankers; scan 16.75 µs vs index 1.49 µs; identical results everywhere ("YES"). Without synonyms Recall@3 drops to 0.794.*

```
------------------------------------------------------------------
Scaling (runtime only, replicated corpus, bm25f, k = 3, median over all 20 queries)
passages  build ms   V       postings  index KB    scan us     index us    speed-up
155       0.162      670     1501      64.0        15.00       1.48        10.1x
1550      0.994      679     16405     238.7       159.27      12.56       12.7x
7750      3.372      719     82645     953.9       841.08      61.57       13.7x
15500     6.561      769     165445    1868.1      1900.52     129.95      14.6x

------------------------------------------------------------------
Top-k selection: bounded heap vs merge sort of all candidates (index engine, bm25f, broad query,
minCoverage 0 so every candidate is ranked)
  15500 passages, C = 9700 candidates, k = 3   : heap 679.3 us, merge sort 2536.4 us, same top-k: yes
  15500 passages, C = 9700 candidates, k = 10  : heap 680.3 us, merge sort 2533.0 us, same top-k: yes
  15500 passages, C = 9700 candidates, k = 100 : heap 750.9 us, merge sort 2533.1 us, same top-k: yes

CSV files written to results/ (query_results.csv, latency.csv, scaling.csv)
```

Figure 15. *Scaling to 15,500 passages: the speed-up grows from 10.1× to 14.6×. Bounded heap vs merge sort with 9,700 candidates: 679 µs vs 2,536 µs, same top-k.*

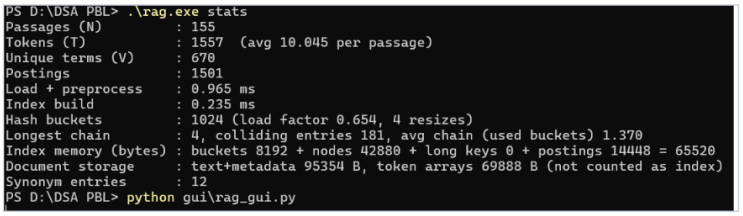


Figure 16. *rag stats: 155 passages, 1,557 tokens, 670 terms, 1,501 postings; index built in 0.235 ms; 65,520 bytes of index memory.*

6. Program output: desktop GUI

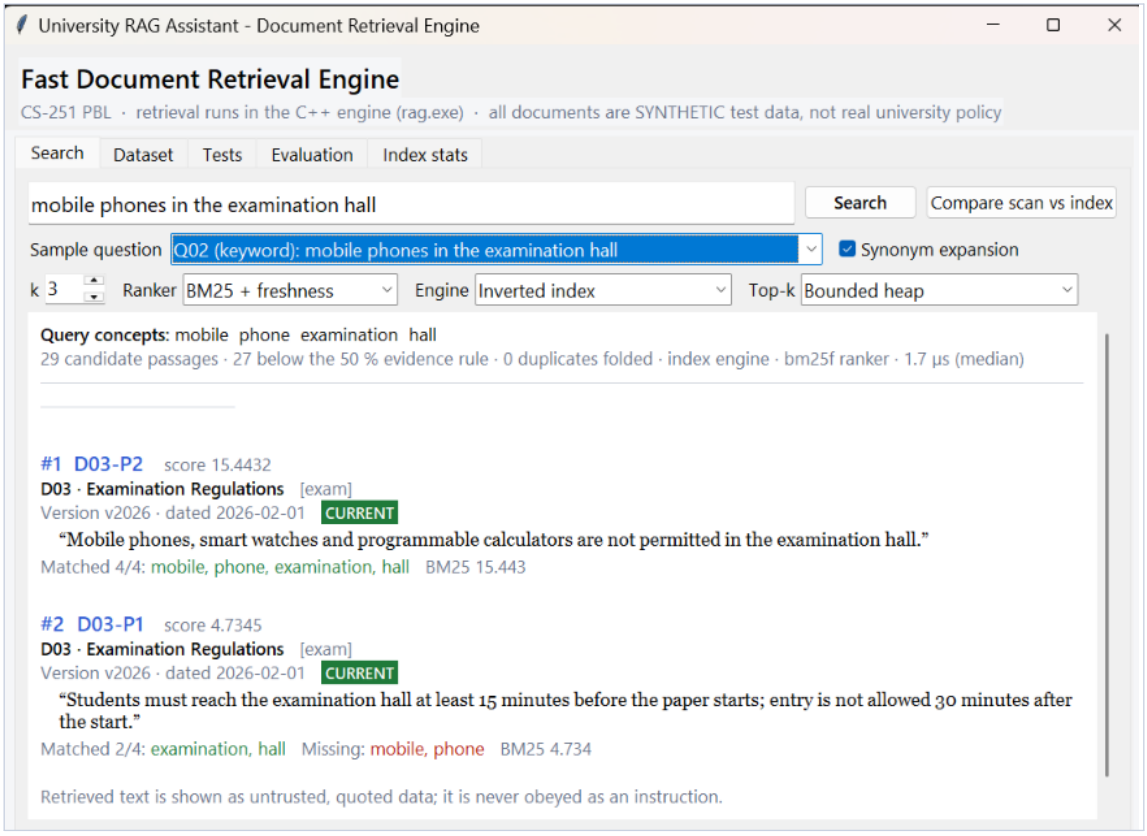


Figure 17. *Search tab: a sample question with provenance, CURRENT badge, and matched/missing concepts.*

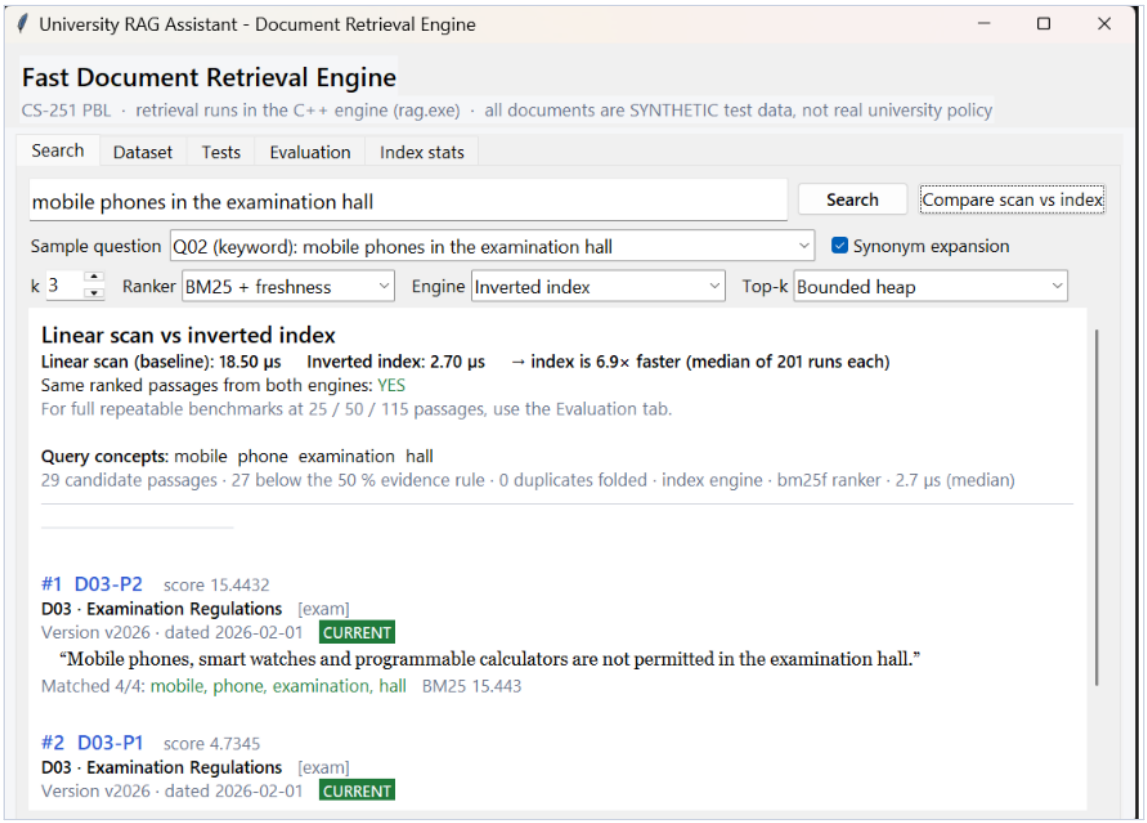
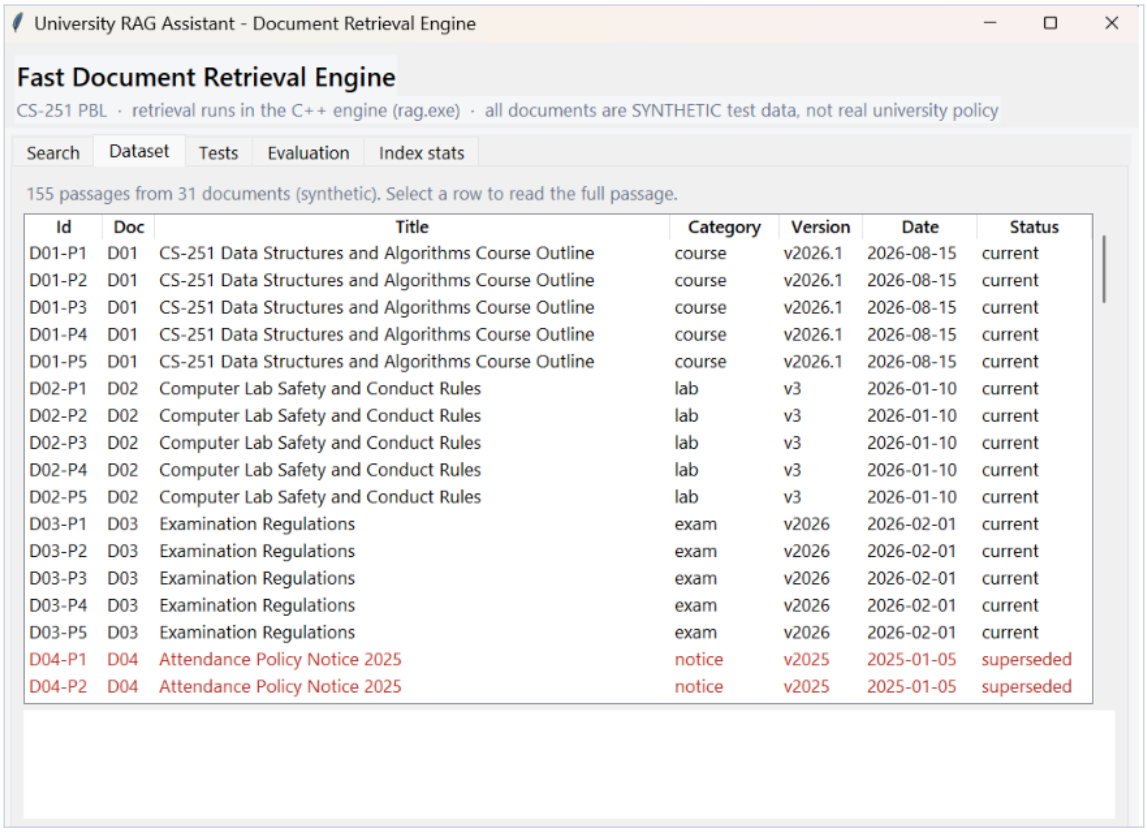


Figure 18. *Compare scan vs index (median of 201 runs each): the index is 6.9× faster and both engines return the same ranked passages.*



Id	Doc	Title	Category	Version	Date	Status
D01-P1	D01	CS-251 Data Structures and Algorithms Course Outline	course	v2026.1	2026-08-15	current
D01-P2	D01	CS-251 Data Structures and Algorithms Course Outline	course	v2026.1	2026-08-15	current
D01-P3	D01	CS-251 Data Structures and Algorithms Course Outline	course	v2026.1	2026-08-15	current
D01-P4	D01	CS-251 Data Structures and Algorithms Course Outline	course	v2026.1	2026-08-15	current
D01-P5	D01	CS-251 Data Structures and Algorithms Course Outline	course	v2026.1	2026-08-15	current
D02-P1	D02	Computer Lab Safety and Conduct Rules	lab	v3	2026-01-10	current
D02-P2	D02	Computer Lab Safety and Conduct Rules	lab	v3	2026-01-10	current
D02-P3	D02	Computer Lab Safety and Conduct Rules	lab	v3	2026-01-10	current
D02-P4	D02	Computer Lab Safety and Conduct Rules	lab	v3	2026-01-10	current
D02-P5	D02	Computer Lab Safety and Conduct Rules	lab	v3	2026-01-10	current
D03-P1	D03	Examination Regulations	exam	v2026	2026-02-01	current
D03-P2	D03	Examination Regulations	exam	v2026	2026-02-01	current
D03-P3	D03	Examination Regulations	exam	v2026	2026-02-01	current
D03-P4	D03	Examination Regulations	exam	v2026	2026-02-01	current
D03-P5	D03	Examination Regulations	exam	v2026	2026-02-01	current
D04-P1	D04	Attendance Policy Notice 2025	notice	v2025	2025-01-05	superseded
D04-P2	D04	Attendance Policy Notice 2025	notice	v2025	2025-01-05	superseded

Figure 19. *Dataset tab: all 155 passages with metadata; superseded passages are shown in red.*

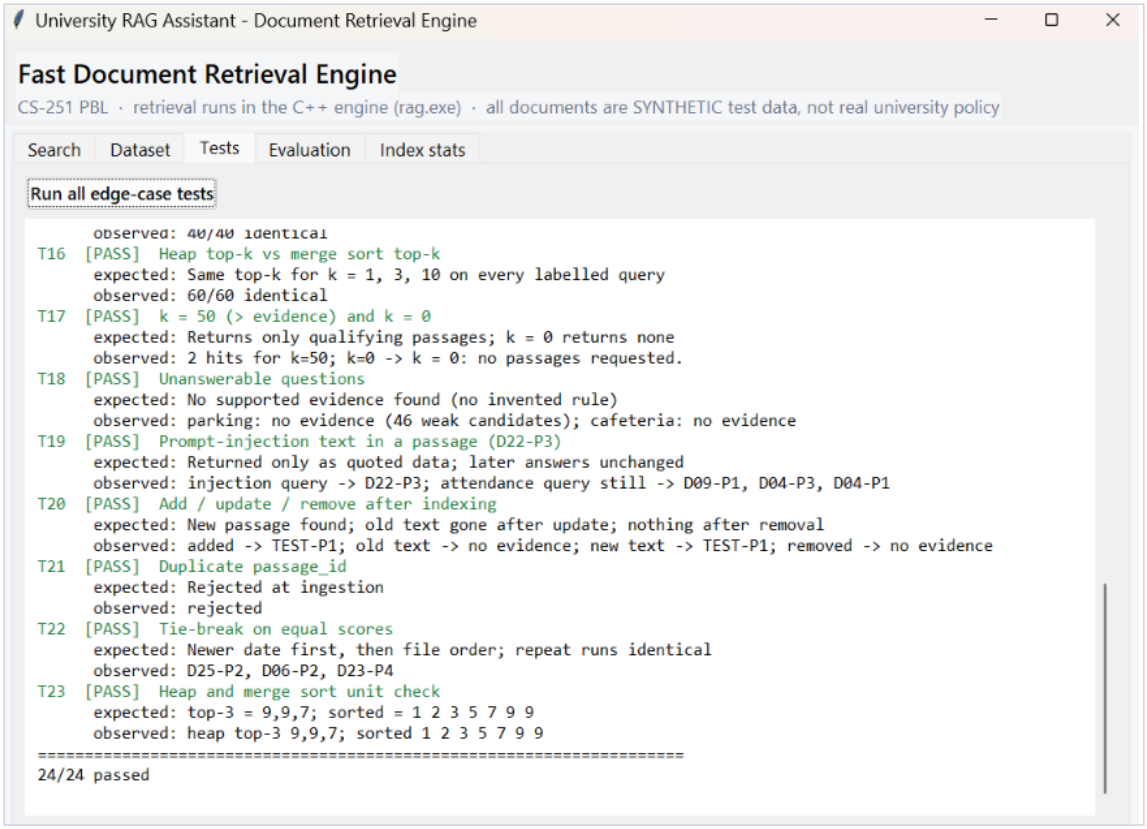


Figure 20. *Tests tab: the same 24 tests, 24/24 passed.*

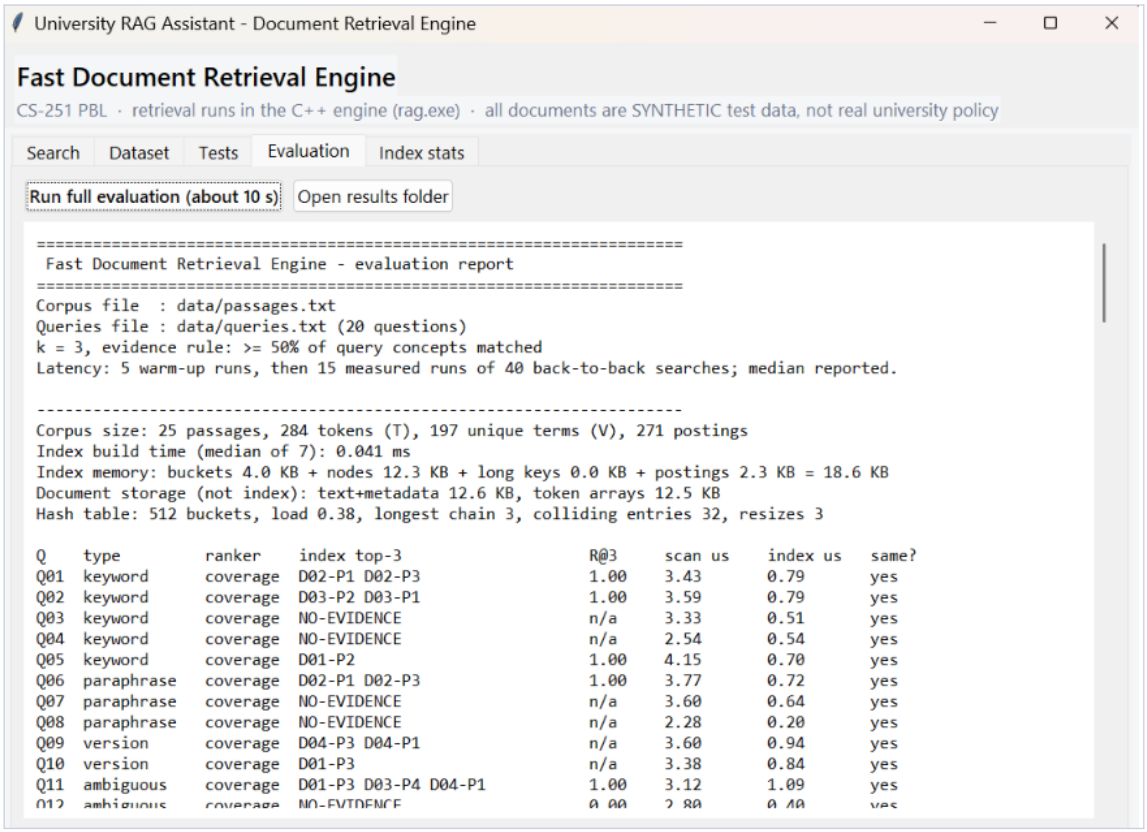


Figure 21. *Evaluation tab: runs rag eval inside the GUI.*

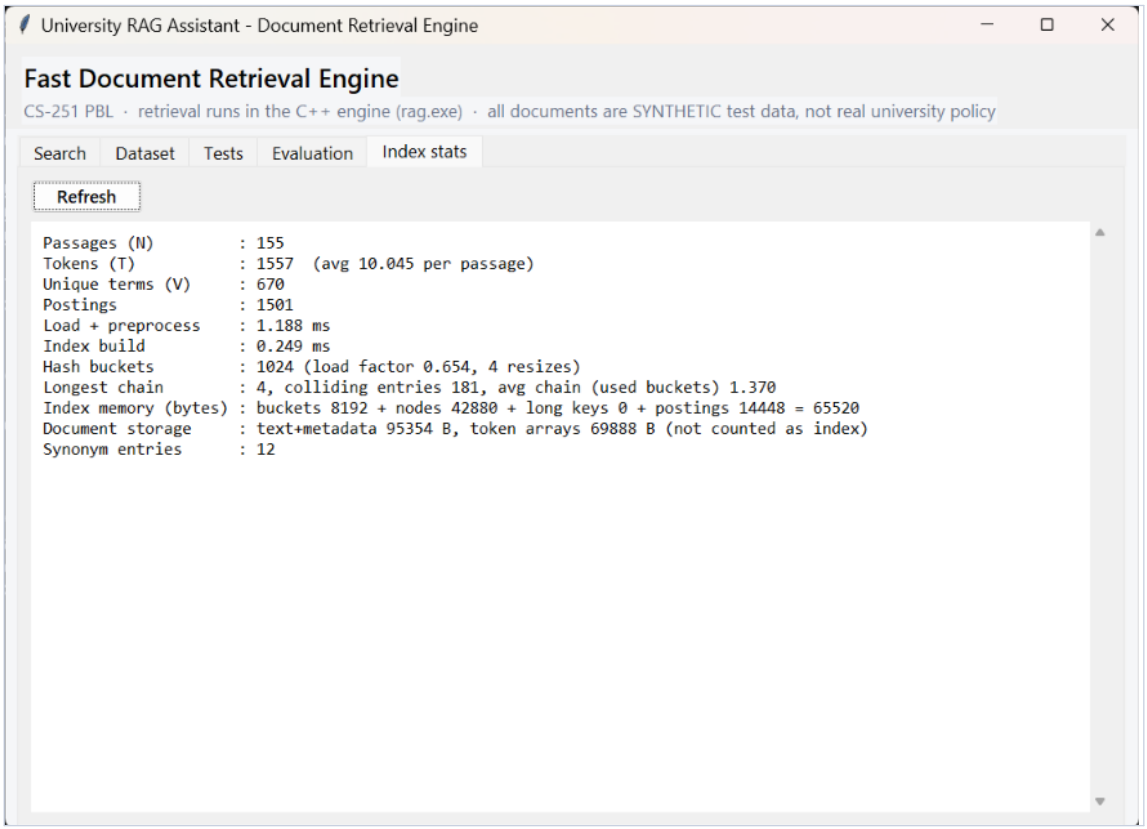


Figure 22. *Index stats tab: vocabulary, hash-table health and memory figures.*

7. Conclusion

- The inverted index returns **exactly the same answers** as the linear scan, and is about **11× faster** at 155 passages and **15×** at 15,500. The gap grows with corpus size.

- The **freshness rule** puts current notices above superseded ones; the **evidence rule** makes the program refuse unanswerable questions instead of guessing.
- **Limitations:**
  - Keyword search misses paraphrases without synonyms (0.794).
  - Near-duplicates are not folded.
  - Version status is entered by hand.
  - The MUET documents are synthetic.
- Timings differ slightly from run to run (about ±10 %); the trends are stable.